

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458511.8808 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.027
Transit depth (Rp/Rs)^2	2.11 +/- 0.78 [%]
Orbital Inclination (inc)	80.74 +/- 1.65
Airmass coefficient 1 (a1)	0.8325 +/- 0.0077
Airmass coefficient 2 (a2)	0.13 +/- 0.04
Scatter in the residuals of the lightcurve fit is	0.92 %
Best Comparison Star	#2 - [153, 47]
Optimal Aperture	4.54
Optimal Annulus	12.33
Transit Duration (day)	0.044 +/- 0.0078

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.027 vs 0.1594 (deviation 0.53σ)
Duration fit vs published	0.044 d vs 0.0512 d (deviation 0.93σ)
Mid-time O–C	3.9 min
Depth SNR	5.4
Residual scatter	0.92 %

Light curve

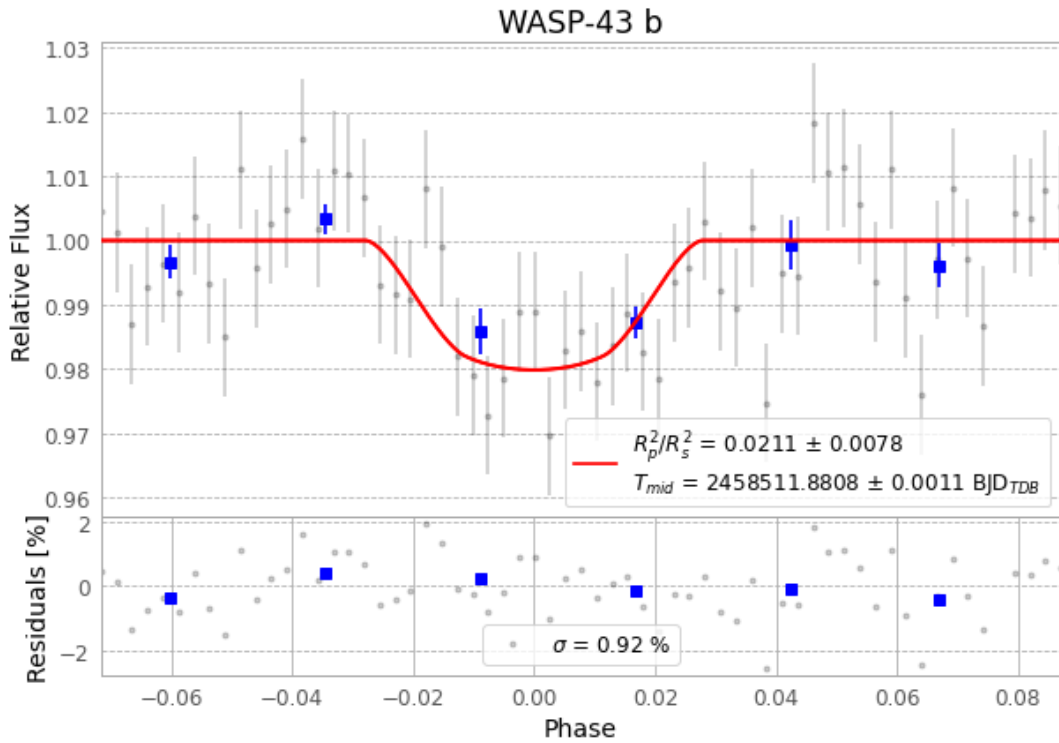


Figure 1 — FinalLightCurve_WASP-43 b_2019-01-28.png (SHA-256 5facaf6dfec8000d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [449, 170], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 32a1cd6478e125a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

63 FITS frames (41 MB), WASP-43190128073612.FITS → WASP-43190128104212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-190128.log (d1a16b86a581...), FinalLightCurve_WASP-43 b_2019-01-28.png (5facaf6dfec8...), FinalLightCurve_WASP-43 b_2019-01-28.csv (e29ae68d8cb6...)

Compute resources

Wall time 112.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 00:25Z.